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BHARAT PROCESS INDUSTRIES PVT. LTD.
NEAR MISS & OBSERVATION REPORTS – UNIT 3
Compiled: 2022 and 2023 Records

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NEAR MISS REPORT NM-2022-008

Date : 14-July-2022
Location : Building C, Ground Floor
Equipment : Pump P-104
Reported By : Prakash Tiwari (Operator)
Supervisor : Suresh Nair

Description:

During monsoon season, excessive humidity in Building C caused condensation on the motor terminal box of Pump P-104. The terminal box cover was found partially open during morning rounds. No electrical fault occurred but the potential for short circuit was high. Cover secured and sealed with silicone.

Root Cause: Terminal box cover latch was corroded and not latching properly. Replaced latch.

Corrective Action: Inspect all motor terminal boxes in Unit 3 for latch condition before every monsoon season (June).

Closed: Yes | Action Verified: Yes | Date: 20-July-2022

NEAR MISS REPORT NM-2022-014

Date : 03-September-2022
Location : Building C, Ground Floor
Equipment : Pump P-105
Reported By : Sanjay Patil (Operator)
Supervisor : Vikram Reddy

Description:

During routine changeover from P-104 to P-105, P-105 started but immediately tripped on high current (58A vs limit 52A). Investigation found discharge valve was opened too quickly (within 5 seconds vs procedure requirement of 30-60 seconds per SOP-CW-003 Section 4.2 Step 3). No damage occurred.

Root Cause: Operator error – new operator (Sanjay Patil, 3 months experience) not fully familiar with SOP requirement on slow valve opening.

Corrective Action: Refresher training for all operators on SOP-CW-003. Valve opening time requirement to be added to DCS operator guidance screen.

Closed: Yes | Action Verified: Partial | Date: 10-Sep-2022
NOTE: DCS guidance screen update still pending as of Jan 2024.

INCIDENT REPORT IR-2022-019

Date : 22-September-2022
Location : Building D, Level 1
Equipment : V-22 (Steam Pressure Vessel 22)
Reported By : Amit Sharma (Technician)
Supervisor : Vikram Reddy

Description:

Safety relief valve SRV-22A lifted spontaneously during normal operation at working pressure of 21.8 bar. Set pressure was 24.8 bar – lifting at 21.8 bar indicates valve set pressure had drifted significantly. Steam release lasted 4 minutes. No injury. Area evacuated per EEP-001.

Root Cause: SRV spring fatigue. Valve had been in service for 5 years without replacement. OEM (Spirax Sarco) recommends spring replacement every 3 years for this service.

Corrective Action:

1. SRV-22A spring replaced and recalibrated.
2. All SRVs in plant to be audited for last spring replacement date.
3. SRV spring replacement added to PM schedule (3-year interval).

Closed: Yes | Action Verified: Yes | Date: 30-Sep-2022

NEAR MISS REPORT NM-2023-003

Date : 18-July-2023
Location : Building C, Ground Floor
Equipment : Pump P-104
Reported By : Ramesh Yadav (Operator)
Supervisor : Suresh Nair

Description:

Second monsoon season observation (see also NM-2022-008): Despite latch repair in July 2022, the terminal box cover of P-104 motor was again found loosely closed. Investigation found the silicone sealant from 2022 had degraded. Additionally, the motor ventilation louvers had insect nesting debris blocking 30% of airflow, contributing to higher motor temperature (motor casing 72°C vs normal 55-60°C).

NOTE: This is the SAME equipment, SAME location, SAME season (monsoon July) as NM-2022-008. Pattern is emerging.

Root Cause: Seasonal – monsoon humidity causes seal degradation. Insect nesting in ventilation is a recurring monsoon problem.

Corrective Action:

1. Terminal box resealed with better grade silicone (Dow Corning 732).
2. Motor ventilation inspected and cleaned.
3. Pre-monsoon inspection checklist (June each year) to be created specifically covering: terminal box seal, motor ventilation, any outdoor/semi-outdoor equipment.

Closed: Yes | Action Verified: Partial | Date: 25-July-2023

NOTE: Pre-monsoon checklist was drafted but not formally approved.

NEAR MISS REPORT NM-2023-011

Date : 12-October-2023
Location : Building C, Ground Floor
Equipment : MV-31 (Motor Control Valve)
Reported By : Deepak Joshi (Technician)
Supervisor : Suresh Nair

Description:

During periodic valve stroke test, MV-31 (cooling water minimum flow bypass) was found stuck at 65% open – not achieving full

stroke. This valve is critical for pump protection. If pump runs with discharge valve closed, MV-31 ensures minimum flow to prevent overheating. A stuck valve could lead to pump damage or mechanical seal failure.

Root Cause: Actuator limit switch had shifted due to vibration from nearby pump (P-104). This is the same root cause as WO-2024-011 (March 2024) where the same fault recurred.

Corrective Action (2023): Limit switch realigned.
NOTE: Corrective action from 2023 did NOT include lock-wiring the limit switch – same fault recurred in March 2024 (see WO-2024-011). Lock wire was only added in March 2024 repair.

Closed: Yes | Action Verified: Yes | Date: 18-Oct-2023

NEAR MISS REPORT NM-2023-018

Date : 28-November-2023
Location : Building C, Level 2
Equipment : HX-11 (Heat Exchanger 11)
Reported By : Deepak Joshi (Technician)
Supervisor : Vikram Reddy

Description:

During inspection of HX-11, technician noticed a small pinhole leak on the tube-side nozzle flange. Cooling water was dripping at approximately 20 ml/min. Leak was in an area that could contact hot surfaces on adjacent equipment.

No fire or injury. Pump was isolated and flange gasket replaced as emergency corrective action.

Root Cause: Gasket degradation – original gasket was spiralwound CS/graphite. For cooling water service with chemical dosing, Thermax (OEM) recommends PTFE envelope gasket. Wrong gasket material had been installed during previous maintenance (unknown date – no record found).

Corrective Action:

1. Replaced with correct PTFE envelope gasket (OEM recommended).
2. Check material specification on all HX flanges during next scheduled maintenance.

Closed: Yes | Action Verified: Yes | Date: 05-Dec-2023

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PATTERN ANALYSIS NOTE (Added by Vikram Reddy, 20-April-2024)

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On reviewing the above records for the IR-2024-007 investigation, the following patterns are visible:

PATTERN 1 – MONSOON VULNERABILITY (July-September):

NM-2022-008 (Jul-2022), NM-2023-003 (Jul-2023) – same equipment (P-104), same season, same basic cause (moisture ingress/motor issues). Pre-monsoon inspection checklist was drafted but never formally approved. Risk of NM-2024 in Jul-2024 is HIGH if the checklist is not activated.
ACTION: Approve and implement pre-monsoon checklist before June 2024.

PATTERN 2 – CORRECTIVE ACTIONS NOT SUSTAINED:

NM-2022-014 → DCS screen update never completed (Jan 2024 note)
NM-2023-011 → Lock wire not added in 2023 → same fault March 2024

IR-2021-003 → Lubrication interval corrected but later reset to
wrong value → same bearing failure March 2024 (IR-2024-007)
SYSTEMIC ISSUE: No verification system exists to confirm long-term
corrective action effectiveness. Actions are "closed" too quickly
without follow-up verification after 3-6 months.

PATTERN 3 – PRE-ALARM TREND IGNORED:

WO-2024-001 (Jan 2024) noted bearing temp at 72°C on P-104.
Operator Round Sheets show temp rising from 2-March onwards.
No supervisor review triggered until alarm at 75°C on 2-March.
RCA on IR-2024-007 identified same pattern.
RECOMMENDATION: Implement trend-based alerting, not just
threshold-based alarming.

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